The Research of Feature Extraction Based on MFCC for Speaker Recognition

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***Abstract*—The feature extraction has proved to a primary issue of speaker recognition that represent the personality of the speaker from speech signals. In the paper, a new approach is presented for speaker recognition using the improved Mel frequency cepstral coefficients (MFCC).The experimental database consists of 30 speakers, 15 male and 15 female, collected in a sound proof room. The result of this experiment certificates that the improved Mel frequency cepstral coefficients derived parameters perform better than traditional Mel frequency cepstral coefficients based on hidden Markov models.**

Keywords—speaker recognition; Mel frequency cepstral coefficients; feature extraction; weighted Mel frequency cepstral coefficients

I. INTRODUCTION

Speech signal processing technology is an indispensable technology in the information society, and speaker recognition is an important research field of speech processing. Speaker recognition is also called the voiceprint recognition, which makes it possible to identify or verify the identity of the speaker using the speech feature. It combines the theories of various subjects, such as acoustics, phonetics, linguistics, physiology, digital signal processing, pattern recognition and artificial intelligence etc. [1].Speaker recognition has a wide application prospect in the judicial identification, security monitoring, e-commerce and other fields. The extraction of the Mel frequency cepstral coefficients is one of the popular approaches of feature extraction [2].

II. FEATURE SELECTION

The effective information is obtained from the input speech signal in the feature selection and extraction stage. The aim of feature selection and extraction is to reduce the dimensions of extraction vector and represent the speech signal with fewer dimensions [3]. Feature selection and measurement are very important [4,5]. We describe the different feature extraction methods in the paper. These features are represented as feature vectors in the feature space. The feature extraction consists of speech pre-processing and MFCC.

A. Front-End Processing

The task of speech front-end processing is to find some useful information from the input speech signals [6]. Speech front-end processing consists of digitization, promoting frequency, framing, and removing silence.

1) Digitization

In speech signal processing, we need to segment the analog audio signal waveform by sampling. The sampling is to attain an amplitude value from the sound waveform at certain intervals in time and convert the continuous signal to discrete signal. In the processing of quantization, the sampled signal is divided into finite intervals with the amplitude of the whole waveform and the sample quantization value is given by classifying. According to Nyquist Sampling Theory, the sampling frequency is usually between 15 kHz and 20 kHz. The speech signal is sampled at 16 KHz, and it is shown as figure 1.

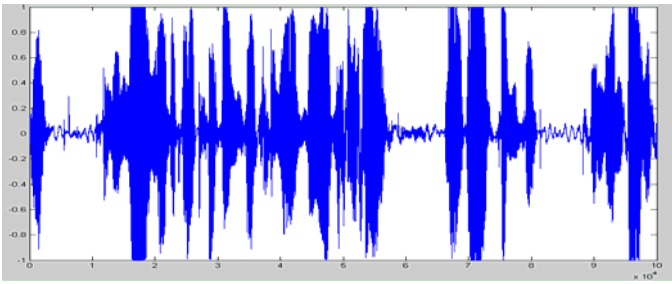

Fig. 1. Speech signal

2) Promoting frequency

In the process of calculating the speech spectrum, the calculation of the high frequencies is more difficult than that of the low frequencies. In order to compensate the high-frequency part, the pre-emphasis must be used in speech pre-processing. Moreover, the importance of high-frequency formants can be also amplified by pre-emphasis. Pre-emphasis is also usually used to promote for the -6dB per octave spectral slope of the speech signal. The output signal $y(n)$ is given by:

$$y(n) = x(n) - ax(n-1) \qquad 0.9 \le a \le 1 \tag{1}$$

The z-transform of the filter is given by:

$$H(z) = 1 - az^{-1} \tag{2}$$

3) Framing

Speech signal is a typical non-stationary signal. Therefore, the speech signal can often be assumed short-time stationary between 10ms to 30ms, and the spectrum and some physical parameters can be approximately regarded as invariant. Typically, the feature extraction is performed on 20 to 30 ms windows with 10 to 15 ms shift between two consecutive windows. The Hamming function is

$$\omega(n) = 0.54 - 0.46\cos(\frac{2\pi n}{N-1}) \tag{3}$$

where N is a number of data in each frame and $n = 0,1,\ldots,N-1$.

4) Removing silence

The aim of endpoint detection is to distinguish the speech and non-speech segments from digital speech signal. The phrase is considered as a crucial part of the speech signal processing. A good endpoint detector can improve the accuracy and speed of a speech recognition system. In order to attain effective results, it is the most important to select appropriate feature [7]. Many experiments have proved that short-time energy and zero-crossing rate are very useful in distinguishing the speech segments. The short-time energy is given by

$$E_n = \sum_{m-\infty}^{\infty} x^2(m)h(n-m) \tag{4}$$

where $h(n)$ is a windowing function. For simplicity a rectangular windowing function is used as defined in

$$\begin{cases} h(n) = 1 & 0 \le n \le N-1 \\ h(n) = 0 & otherwise \end{cases} \tag{5}$$

Zero-crossing rate is he number of zero levels of the speech signal in one frame, and it can reflect the frequency characteristics in a certain extent. We can judge the starting and ending points of the speech using zero crossing rate and short-time energy, namely the endpoint detection [8]. The result of endpoint detection is shown in figure2.A definition for zero-crossing rate is

$$Z_n = \sum_{m=-\infty}^{\infty} |\mathrm{sgn}[x(m)] - \mathrm{sgn}[x(m-1)]|\omega(n-m) \tag{6}$$

where

$$\mathrm{sgn}[x(n)] = \begin{cases} 1 & x(n) \ge 0 \\ -1 & x(n) \prec 0 \end{cases} \tag{7}$$

and

$$\omega(n) = \begin{cases} \frac{1}{2N} & 0 \le n \le N-1 \\ 0 & otherwise \end{cases} \tag{8}$$

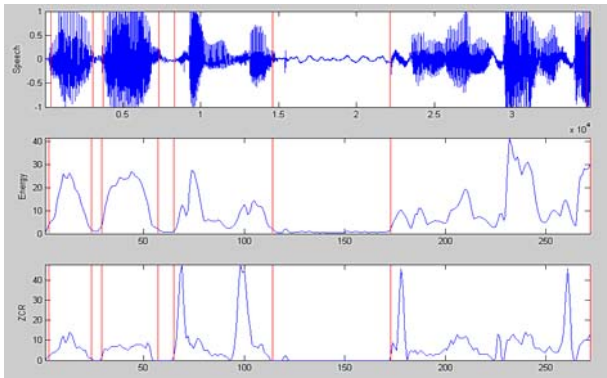

Fig. 2. Endpoint detection

B. MFCC

MFCC represents the ear model, and we can get good results in speaker recognition especially when a high number of coefficients are used [9].

We have extracted the first two stages of MFCC. Its algorithm involves framing, hamming windowing, Fast Fourier Transform, Mel filters, energies log and Discrete Cosine Transform. We can get the magnitude spectrum of the windowed speech data by using the FFT algorithm. The Mel-filtering provides a model of hearing realized by the bank of triangular filters uniformly spaced in the scale Mel. The scale Mel is given by

$$f_{mel} = 2595\log_{10}(1 + \frac{f}{700}) \tag{9}$$

where f notes the frequency in Hz. figure3 gives the me Mel filter bank.

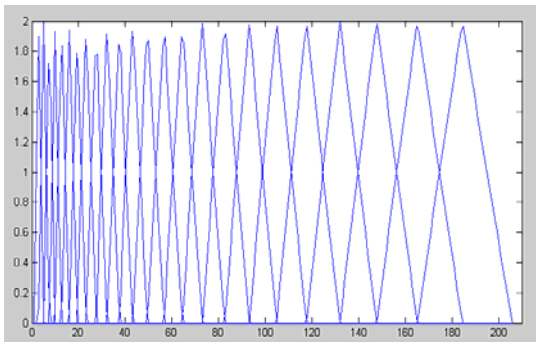

Fig. 3. Mel filter bank

A sequence of finitely many data points in terms of a sum of cosine functions oscillating at different frequencies are represented by a discrete cosine transform (DCT). The last algorithm stage performed to obtain Mel frequency cepstral coefficients is a Discrete Cosine Transform (DCT) which encodes the Mel logarithmic magnitude spectrum to the Mel frequency cepstral coefficients (MFCC). The Mel frequency cepstral coefficients are

$$C_i = \sqrt{\frac{2}{p}} \sum_{j=1}^{p} m_j \cos[\frac{\pi i}{p}(j-0.5)] \tag{10}$$

where p is the number of filters and m_j is the coefficient of filter.

C. Weighted MFCC

The static characteristics of the voice are described by the Mel frequency cepstral coefficients. Changing voice signals are an important feature of speech signal, so we introduce the first order differential MFCC parameters [10].

$$d(n) = \frac{1}{\sqrt{\sum_{i=-k}^{k} i^2}} \sum_{i=-k}^{k} i \times c(n+i) \tag{11}$$

where k is a constant, usually take k =2 . Noise to some extent can be eliminated by the differential MFCC, so we can achieve the better performance.

The first coefficient, which is highly correlated with the energy of the frame, is not kept. From the result of experiments, we get an evidence that different dimension represents different feature of the voice. As the speech signal processing experiences show, the higher order dimensions have a low value, and the noise influences the lower order dimension easily. So we make

$$\hat{c}_i = \omega_i c_i \tag{12}$$

$$\omega_i = \alpha + (1-\alpha)\ X_i \tag{13}$$

$$X_i = x_i / X \tag{14}$$

$$x_i = a e^{-b|i-c|} \sin(\pi i / I) \tag{15}$$

where α , b , c are constants, X is the maximum of x_i , i is the feature dimension, I is the total feature dimension.

ω_i has increased with every increment in the feature dimension of speech signal. When the feature dimension increases to c, the weighted coefficient reaches maximum value. With the continuous increment of the feature dimension, ω_i will start to decrease [11].

III. EXPERIMENT AND RESULT

We recorded the audio in an acoustic room by using the handset microphone. Because the space is limited, we only test our system using a small speech database that consist of 20 men and 20 women speakers. All speakers uttered sentences once in a training stage and once in a testing stage. The recording contains 10 sentences for each speaker, recorded over a period of up to a month.

We sampled the analog speech at a 16 KHz rate, and then pre-emphasized the digital speech signals using a coefficient of 0.97. We derived the features from speech frame of length 25ms with a frame rate of 10ms, and windowed each frame using a Hamming window function. Unvoiced speech samples were removed by endpoint detection. Speech samples were parameterized with MFCC. Each frame of speech was represented by a 30 dimensional feature vector which consists of 15 MFCC with first differentials appended.

The recognition rate for the different number of mixture is shown in figure 4 and the ordinate is the recognition rate (%) and the abscissa is M (the number of mixture). The recognition accuracy using weighted parameters is better than that using non-weighted parameters and M reflects the performance of pattern recognition. The larger M is, the higher recognition performance is get. But M is too much, the storage capacity and computing are too large and speaker recognition performance did not get greatly improved.

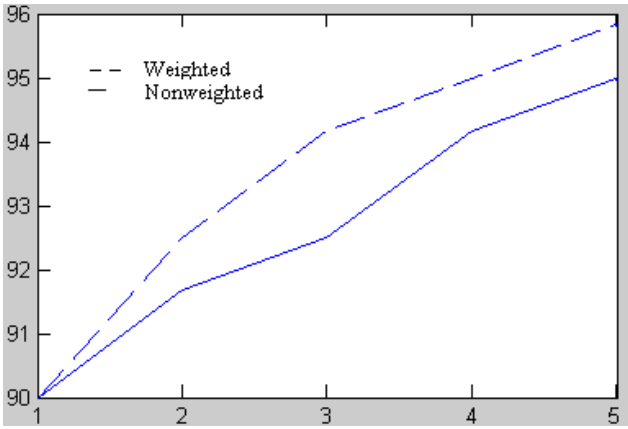

Fig. 4. Recognition rate for the different number of mixture

The recognition rate for various testing time is shown in figure 5 and the ordinate is the recognition rate (%) and the abscissa is time (second). The recognition rate of characteristic parameters for the weighted is superior to non-weighted and testing time reflects the performance of pattern recognition. The longer the testing time is, the higher recognition performance is get. When circumstances permit, we should make the testing time longer. But the testing time is too long, the speed of recognition will slow.

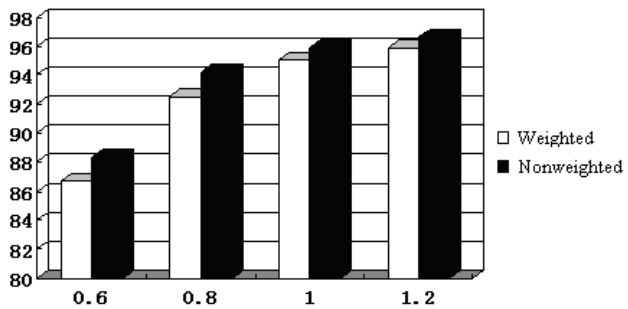

Fig. 5. Recognition rate for various testing time

IV. CONCLUSION

The principle of speaker recognition is described, and a novel method of feature extraction is presented in the paper. It is clear that we find that recognition performance is improved by approximately 1% with weighted MFCC, and we confirmed that our proposed method is superior to conventional methods from experiment data. The novel method reduced the memory

and calculation load. The improved feature can represent the essential characteristics effectively.

As future work, we suggest that the number of feature dimension and the weighting function should be optimized [12].we will try our best to a modified method that can raise the accuracy of the speaker recognition.

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